

OPNsense — Console Menu Configuration

Arch (Dell Inspiron 530) • “Sentry Gate” • OPNsense 26.7 amd64 • single NIC (em0) • console-driven

Rewritten around the console menu you actually have in front of you, rather than web GUI navigation. The console can do everything that matters right now; the GUI items are separated out and deferred. Root password is already done, so that step is dropped.

First: your LAN interface may have no IP address

The console header shows LAN (em0) -> with nothing after the arrow. A configured interface normally reads LAN (em0) -> v4: 192.168.10.1/24.

If that is accurate and not just cut off in the photo, em0 has no address — which means no web GUI, no reliable connectivity, nothing. **Fix this first with option 2.**

This also still holds: do not connect Arch to the family network until the DHCP server is confirmed off and the subnet is off 192.168.1.x. Option 2 handles both in one pass.

What each menu option is for

#	Option	What it does / when you need it
0	Logout	Drops back to the login prompt. Use it when walking away from the machine.
1	Assign interfaces	Re-runs interface assignment. This is the one you will need when the second NIC arrives — it is where you tell OPNsense which physical port is WAN and which is LAN.
2	Set interface IP address	The workhorse right now. Sets the address, subnet, gateway, and — importantly — asks whether to run a DHCP server. See the walkthrough below.
3	Reset the root password	Already done. Keep in mind it exists as a recovery path if you ever lock yourself out.
4	Reset to factory defaults	The panic button. Wipes config back to stock. Cheap to use at this stage since you have almost nothing configured — much more expensive later.
5	Power off system	Use this rather than pulling power. See the gotcha on USB media below.
6	Reboot system	Clean restart. Same reasoning as above.
7	Ping host	Quick connectivity test without dropping to a shell. Useful for confirming a gateway is reachable.

#	Option	What it does / when you need it
8	Shell	Full FreeBSD root shell. Used below for verification and config backup. Type <code>exit</code> to return to this menu.
9	pfTop	Live view of active firewall states and connections. Not useful yet, very useful once traffic is flowing through.
1 0	Firewall log	Live tail of what the firewall is passing and blocking. Your primary debugging tool once rules exist.
1 1	Reload all services	Restarts services without a full reboot. Try this before rebooting when something seems stuck.
1 2	Update from console	Firmware updates without the GUI. Requires working internet, so it is gated on having a WAN — see the note on this below.
1 3	Restore a backup	Restores a saved config. Pairs with the backup step below; this is your fast rebuild path.

Step 1 — Option 2: set the LAN address and kill DHCP

This single option covers what were three separate GUI tasks in the previous version. It is an interactive prompt sequence; the answers you want are shown after each. Wording varies slightly between versions, so match on meaning rather than exact phrasing.

```

Enter an option: 2

Available interfaces:
  1 - LAN (em0)
Enter the number of the interface to configure:  -> 1

Configure IPv4 address LAN interface via DHCP? (y/n)  -> n

Enter the new LAN IPv4 address:                      -> 192.168.10.1
  (anything OFF 192.168.1.x; 10.10.10.1 also fine)

Enter the new LAN IPv4 subnet bit count (1 to 32):    -> 24

For a LAN, press <ENTER> for none (gateway):          -> just press ENTER

Configure IPv6 via WAN tracking? (y/n)                -> n
Configure IPv6 via DHCP6? (y/n)                      -> n
Enter the new LAN IPv6 address:                      -> just press ENTER

Do you want to enable the DHCP server on LAN? (y/n)  -> n   <<< IMPORTANT

Change web GUI protocol from HTTPS to HTTP? (y/n)    -> n
Generate a new self-signed web GUI certificate? (y/n) -> n
Restore web GUI access defaults? (y/n)               -> n

```

When it finishes, the header at the top of the menu should now read something like LAN (em0) -> v4: 192.168.10.1/24. That line is your confirmation — if it is still blank, something did not take.

Reaching the GUI after this

Set your laptop's Ethernet adapter to a static address on the same subnet — e.g. 192.168.10.50, mask 255.255.255.0, no gateway — since you just turned off DHCP and nothing will hand you an address.

Then browse to `https://192.168.10.1`. You will get a certificate warning; that is expected on a self-signed cert. The console header prints the cert's SHA256 fingerprint — you can compare it against what the browser shows if you want to verify you are talking to the right box.

Step 2 — Option 8: verify and back up from the shell

Drop to the shell with option 8. These four checks cover what the GUI version of this guide asked you to verify, and the backup is the important one. Type `exit` when finished to return to the menu.

```
# 1. Confirm the interface actually took the address
ifconfig em0

# 2. Confirm RAM disks are active (protects the USB stick from log writes)
mount | grep -E '/tmp|/var'
#   Look for /dev/md0, /dev/md1 etc. -- 'md' means memory disk = good.
#   If you see the USB device instead, RAM disks are OFF; enable them
#   later in the GUI under System > Settings > Miscellaneous.

# 3. Check free space on the boot media
df -h /

# 4. Back up the running config -- do this every time you change something
cp /conf/config.xml /conf/config-backup-$(date +%Y%m%d).xml
ls -la /conf/

exit
```

Get the backup off the box, not just onto it

A copy sitting on the same USB stick does not survive that USB stick failing, which is the failure mode most likely to actually happen here.

Once you have GUI access, use System → Configuration → Backups to download the XML to your laptop. Until then, the in-place copy above is better than nothing.

Recovery path if the stick dies: reflash the nano image, then use console option 13 to restore that XML. Roughly ten minutes instead of redoing everything.

Step 3 — Option 12: firmware updates (optional, needs internet)

With one NIC assigned as LAN, Arch has no internet, so option 12 will fail. There is a safe workaround if you want to update now rather than waiting on the PCIe card:

- Use option **1) Assign interfaces** to temporarily assign `em0` as **WAN** instead of LAN.

- Plug Arch into the family router. This is safe in this configuration specifically — a WAN interface runs a DHCP *client*, not a server, so there is no rogue-DHCP risk to the family network.
- Run option **12) Update from console** and let it pull patches.
- Use option 1 again to put em0 back to LAN, then option 2 to re-set the static address.

You can only do this because you have a working VGA console. With a WAN-only assignment the GUI is unreachable by default, so the console is your only way back. Worth doing while the console is proven to work.

Deferred — genuinely needs the GUI or the second NIC

None of these are urgent, and none of them are reachable from the console menu. Do not burn time trying.

- **Hostname and domain** (currently the default `OPNsense.internal`) — cosmetic, GUI only, no rush.
- **Timezone and NTP** — set to America/Chicago. Matters for Wazuh log correlation later, not now.
- **Web GUI hardening** — HTTPS-only, non-default port, SSH key-only.
- **WAN/LAN split** — option 1, once the PCIe NIC is physically installed.
- **Firewall rules, default-deny** — meaningless with one interface; there is nothing to filter between.
- **Suricata** — install once WAN exists. Run in IDS/alert-only mode for about a week before letting it block, or you will break your own traffic and not know which rule did it.
- **Tailscale subnet router** — the last piece, once the server is actually behind Arch.

Gotchas

Always shut down with option 5

Pulling power from a running system on USB media is the fastest way to corrupt the filesystem. This is a fifteen-year-old machine that has already handed you a failing hard drive and two dead flash drives; do not add an avoidable failure to the list. Option 5 to power off, option 6 to reboot.

The shell is real root

Option 8 drops you to an unrestricted FreeBSD root shell. Fine for the read-only checks above. Avoid hand-editing `/conf/config.xml` while the system is running — OPNsense holds state in memory and will overwrite your edits on the next config write.

You are on 26.7

Newer than most tutorials you will find online, which mostly target 24.x and 25.x. Menu option numbers and prompt wording occasionally shift between releases. If a guide's steps do not match your screen, trust your screen.

Nothing here is load-bearing yet

No family data, no server traffic, no dependencies. This is the cheapest possible moment to break something and use option 4 to start over. That stops being true the moment the server sits behind it.

Sentry Gate build • console-driven revision • replaces the GUI-path version for the single-NIC phase.